

COURSE TITLE : TECHNICAL TEXTILES
COURSE CODE : 5065
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Geo Textiles	13
2	Filtration and Agriculture Textiles	13
3	Hygiene and Medical Textiles	13
4	Industrial Textiles	13
TOTAL		52

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Understand various application & technical features of geo textiles.
2	1	Know the clear idea about various Filter fabrics & agriculture fabrics.
3	1	Understand different Hygiene and Medical Textiles
4	1	Know the various Industrial Textiles

SPECIFIC OUTCOME

MODULE I TECHNICAL TEXTILES

1.1.0 Understand various application of technical Textiles & features of geo textiles

- 1.1.1 Definition and scope of Technical Textiles
- 1.1.2 State various applications of Technical Textiles and fibers used in civil and Agriculture engineering, Filters, Automotive, Medical and Hygiene applications, Protective , defense & industrial applications.
- 1.1.3 State the Basic functions of geo textiles and Essential properties.
- 1.1.4 Classification of geo textiles
- 1.1.5 Explain the Composites and their uses
- 1.1.6 Describe the area of application of geo textiles

MODULE II FILTERS AND AGRICULTURE TEXTILES

2.1.0 Know the clear idea about various Filter fabrics & agriculture fabrics.

- 2.1.1 Describe the Dust filtration fabrics
- 2.1.2 Discuss the General protective mask & high temperature filtration
- 2.1.3 Describe the Purification & separation of Gases and Liquids
- 2.1.4 Explain the Areas of applications of agriculture fabrics.

MODULE III HYGIENE AND MEDICAL TEXTILES

3.1.0 Understand different Hygiene And Medical Textiles

- 3.1.1 Illustrate the essential properties of medical cotton & cloth
- 3.1.2 State the types of suture threads.
- 3.1.3 Describe the dental application of textiles
- 3.1.4 Describe the Properties required for operation theatre clothing
- 3.1.5 Illustrate the requirements of fibres used for Cardio- vascular textiles and Artificial blood vessels

MODULE IV INDUSTRIAL TEXTILES

4.1.0 Know the various Industrial Textiles

- 4.1.1 State the various applications of industrial textiles.
- 4.1.2 State the requirements of materials used in Tyre cords
- 4.1.3 List the physical and mechanical requirements of tyre cord fabrics
- 4.1.4 Illustrate fire hoses
- 4.1.5 State the importance of reinforcement and its constructions ,
- 4.1.6 Explain the materials used for belts
- 4.1.7 Describe the Characteristics of Conveyor belts and transmission belts.
- 4.1.8 Explain the properties required for Bullet proof fabrics.
- 4.1.9 State the properties required for Fire proof fabrics

CONTENT DETAILS

MODULE I GEO TEXTILES

Definition and scope of Technical Textiles - State various applications of Technical Textiles - fibers used - civil and Agriculture Engineering - Automotive and Aeronautics - Medical and Hygiene applications - Protection and defense applications - Basic functions of geo textiles–Essential properties of Geo Textiles -Types of geo textiles - Classification–Woven, Non-woven, Knitted & Biodegradable geo textiles – advantages - materials used - composites and their uses. Geo textile requirements for separation, filtration, drainage, reinforcement, protection & water proofing - geo textiles in temporary and permanent road construction - railway stabilization - fibres and fabrics used for above application.

MODULE II FILTERS AND AGRICULTURE TEXTILES

Introduction- Dust filtration - general, protective masks and high temperature filtration, purification and separation of gases - cigarette filters - liquid filtration - solid liquid filtration, liquid- liquid separation - Textiles used for the above applications and their features. Brief study of textiles for seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for these applications. Tarpaulins - requirement for agricultural purposes.

MODULE III HYGIENE AND MEDICAL TEXTILES

Processes involved in the manufacture of surgical cotton and surgical cloth- materials used- Essential properties of surgical cotton and cloth – suture thread - natural and synthetic - different types used - dental application- uses and properties required for operation theatre clothing like theatre drapes, gowns, masks, surgical covers, sanitary pads – characters required - materials employed - Requirements and fibres used for cardio vascular textiles - woven, knitted - requirements of artificial blood vessels.

MODULE IV INDUSTRIAL TEXTILES

Various applications of industrial textiles -Tyre cords - requirements - materials used - flow chart for tyre cord manufacturing – physical and mechanical requirements of tyre fabrics. Hoses - definition - importance of reinforcement and its constructions - Basic types of reinforcements, fibre requirements for industrial hoses and different types of industrial hoses. Belts - materials used – characteristics of conveyor belts - Transmission belts - flat belt –V belts. Materials used and construction of V-belt.

REFERENCE BOOKS

- 1 A.R.Horrocks & S.C.Anand - Hand book of Technical Textiles - The Textile Institute
- 2 A.A. Vaidya Production of Synthetic Fibres, 1988 - Prentice Hall of India P Ltd., New Delhi
- 3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall
- 4 Pushpa Bajaj& AK Sengupta - Textiles for preparaton and coated fabrics , - Textile progress, Vol.14, No. 1 Textile institute
- 5 R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, Vol.2, 1970 Textile institute
- 6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow
- 7 B S Dhasaradhan - Anatomical Textiles, Surgicals Asian textile Journal June 1993
- 8 B S Dhasaradhan - Hospital& Dialysate textiles Asian textile Journal Dec 1993
- 9 K P Chellaman i - Yarns& Technical Textiles SITRA Publication
- 10 Prof. P A Khatwani& S S Yardi - NCUTE Pilot Programme- Technical textiles NCUTE, IIT, New Delhi